

Knight Path: Supporting Player Sense-making in Game Narratives via Visual Scaffolding and Spoiler-Aware AI

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Abstract

Fragmented narrative delivery in games often causes cognitive gaps, while external resources risk spoilers and information overload. We present Knight Path, an adaptive system integrating structural visual scaffolding with on-demand, spoiler-aware AI guidance. The system combines interactive narrative visualization with a tiered AI assistant synchronized with live gameplay. A user study ($N = 18$) in a non-linear environment showed significant improvements in search efficiency and contextual clarity while preserving narrative immersion. Knight Path effectively addresses the conflict between seeking assistance and sustaining the joy of discovery by empowering players to manage narrative uncertainty.

CCS Concepts

• **Applied computing** → **Computer games**; • **Human-centered computing** → **Interactive systems and tools**; *Visualization*.

Keywords

Games/Play, Creativity Support, Information Seeking & Search

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1 Introduction

In modern video games, narrative is often delivered through indexical (i.e., trace-based) storytelling, where players reconstruct the plot by discovering environmental clues and fragmented dialogues [4]. While this approach encourages curiosity and discovery, the complexity often creates cognitive gaps. Players experience

narrative uncertainty, ranging from minor ambiguity to a subjective impasse that prevents progression without external assistance. Prior research suggests that narrative uncertainty is not discrete or occasional. Instead, it is continuous, such that even minor informational gaps can disrupt comfortable exploration, which in turn induces frustration [3]. To resolve these gaps, players traditionally relied on external resources, such as wikis or walkthroughs [7, 9]. However, these resources introduce a trade-off between the desire to acquire information and preserving discovery with unexpected information, leading to information overload or accidental exposure to spoilers [8]. Recent research on game curiosity suggests that player engagement arises from maintaining a managed level of uncertainty rather than achieving total clarity [14].

Despite these challenges, technological approaches often prioritize analytical utility over narrative discovery. Research on game data visualization has focused on spatio-temporal logs [10] or optimizing tactical performance [15], and even in non-linear genres, the emphasis remains on statistical telemetry rather than supporting narrative interpretation [6]. Frameworks for narrative inference are largely designed for post-hoc analysis by creators or analysts rather than live player support [16]. Consequently, while existing visualizations help structure fragmented information, they offer limited context-sensitive guidance when players encounter narrative uncertainty during live gameplay. More recently, Large Language Models (LLMs) have emerged as a potential solution for providing player assistance [5], yet most rely on proactive intervention that risks interrupting player flow or the *eureka* moment of self-discovery. Furthermore, current LLM assistants fail to synchronize with a player's progression, potentially leading to premature disclosures that compromise the narrative experience [11]. There remains a need for an integrated approach that visually scaffolds fragmented narrative to support memory and comprehension, while providing context-aware guidance only upon explicit player request.

To address these challenges, we present Knight Path, a context-aware narrative support system that integrates structural visual scaffolding with on-demand, spoiler-aware AI guidance to resolve the trade-off in fragmented storytelling. To evaluate the efficacy of the system, *Hollow Knight* was selected as a representative case. Featuring a highly fragmented narrative and a vast, non-linear world, this game serves as an ideal testbed for evaluating narrative support systems—a role it has consistently fulfilled in previous literature

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focusing on story interpretation and visualization [6, 16]. Knight Path integrates a real-time data pipeline with two components: (1) a region-centric narrative mind map that organizes collected clues and (2) a spoiler-aware AI chatbot that provides tiered guidance through four levels (None, Medium, Detailed, High). These components are designed to work in tandem; the mind map enables players to effectively contextualize their current progress, which in turn facilitates the formulation of more precise inquiries to the AI assistant. By adopting a user-initiated interaction model, the system is intended to protect player agency and narrative immersion. The efficacy of Knight Path was evaluated through a user study ($N = 18$) comparing it against conventional web-based searching. The research is guided by two primary questions:

- **RQ1:** How does the proposed system influence players' information seeking and progression processes?
- **RQ2:** How does the proposed system influence players' narrative comprehension experiences?

Our contributions are as follows: (1) a context-aware narrative support system synchronized with live gameplay; (2) a tiered spoiler control mechanism for user-initiated information depth; and (3) empirical insights into how integrated visual graphs and reactive AI influence player behavior and immersion in fragmented narratives.

2 Knight Path

2.1 Design Concept

The development of Knight Path is guided by two core concepts focusing on structural support and player agency.

DC1. Relational Scaffolding for Contextual Understanding: Knight Path utilizes a region-centric narrative mind map to facilitate narrative reconstruction during gameplay. This progression-aware representation integrates the graph-based relational structures of ClueCart [16] and the region-oriented data aggregation of HKViz [6] to support sensemaking in non-linear games. By incrementally constructing nodes as players discover new locations, characters, or items, the system ensures its visual structure remains aligned with their spatial exploration and evolving narrative understanding.

DC2. Pull-based Assistance for Agency Preservation: To preserve player agency and narrative discovery, Knight Path employs a reactive, pull-based interaction model where assistance is provided only upon explicit player request. This design addresses limitations in prior AI assistants that risk disrupting player flow or revealing premature information [5, 11]. By allowing players to control information depth through tiered spoiler levels, the system ensures that AI guidance remains synchronized with individual progression while preserving the joy of self-discovery.

2.2 System Implementation

The visualization interface serves as a dynamic workspace where collected clues are organized into location-based clusters. To realize **DC1**, narrative nodes are generated in real-time during gameplay, ensuring the **narrative mind map** (Figure 1-1) remains synchronized with player progression. Users can directly interact with the visualization, such as by right-clicking nodes to edit memos or double-clicking to revisit dialogue history. To provide contextually

relevant assistance, the AI assistant utilizes a hybrid RAG engine synchronized with individual player states. Visual complexity is managed through **filtering and shortcuts** (Figure 1-2). To realize **DC2**, the **context-aware AI chat** (Figure 1-3) constructs prompts integrating: (1) Level-specific guidelines via the spoiler-level controller; (2) real-time player state (locations, abilities, activities); and (3) Query-based Wiki Knowledge filtered through discovery logs. External knowledge was retrieved from a widely used Korean community wiki [12], restricted to the Hollow Knight entries. The AI responded exclusively in Korean to match both the language of the source material and the participant population. Players can prioritize retrieval by selecting specific nodes. Generated responses trigger automatic highlighting in the visualization (Figure 1-1), bridging text-based guidance with visual context.

The system was implemented as a *React* web application. Real-time logging is handled via a *BepInEx* plugin; in-game events are hooked and stored in *MongoDB* through a *Node.js* backend to generate dynamic visualizations in the web interface. For the RAG engine, external wiki data is embedded using the *Ollama bge-m3* model into a *ChromaDB* vector store, utilizing *GPT-4o-mini* for responses tailored to the selected spoiler level (Details in Appendix A).

3 User Study

3.1 Study Design and Participants

Eighteen university students (mean age 22.5; 16 males, 2 females) who play PC games at least 1-3 times per month were recruited. The **novice group** ($n = 14$) had no prior experience in the game; spoiler sensitivity scores were balanced across the experimental ($n = 7$) and control ($n = 7$) groups to control for individual variance. The **expert group** ($n = 4$) consisted of players who had reached at least one game ending. To evaluate the proposed system, this study employs a between-subjects design for novices. The primary independent variable is the information-seeking environment: the **experimental group**, which was restricted to using *Knight Path*, and the **control group**, which was given unrestricted access to web-based tools, including search engines, YouTube, and existing large language models (e.g., ChatGPT). In addition, an expert group was separately recruited and instructed to use only the proposed system to perform a comprehensive system evaluation and examine qualitative differences in information-seeking strategies and usage patterns between novices and experts. Before the session, all participants received a brief introduction and provided informed consent. The experiment utilized a dual-monitor setup to separate gameplay (primary) and search tools (secondary). Total sessions lasted 150 minutes for novices and 120 minutes for experts, including a 10-minute mid-session break. Reflecting these durations, novices and experts received approximately \$17 and \$14, respectively. Sessions concluded with surveys and semi-structured interviews.

3.2 Measures

Subjective Experience. Subjective measures for **game interest** (pre, post, and post-interest change), **immersion**, and **clarity of context** were evaluated using 5-point Likert scales. Both were adapted from [16]. To quantify the system's support for narrative reconstruction and sense-making, we utilized the **Creativity Support Index (CSI)** [2], and **System Usability Scale (SUS)** for

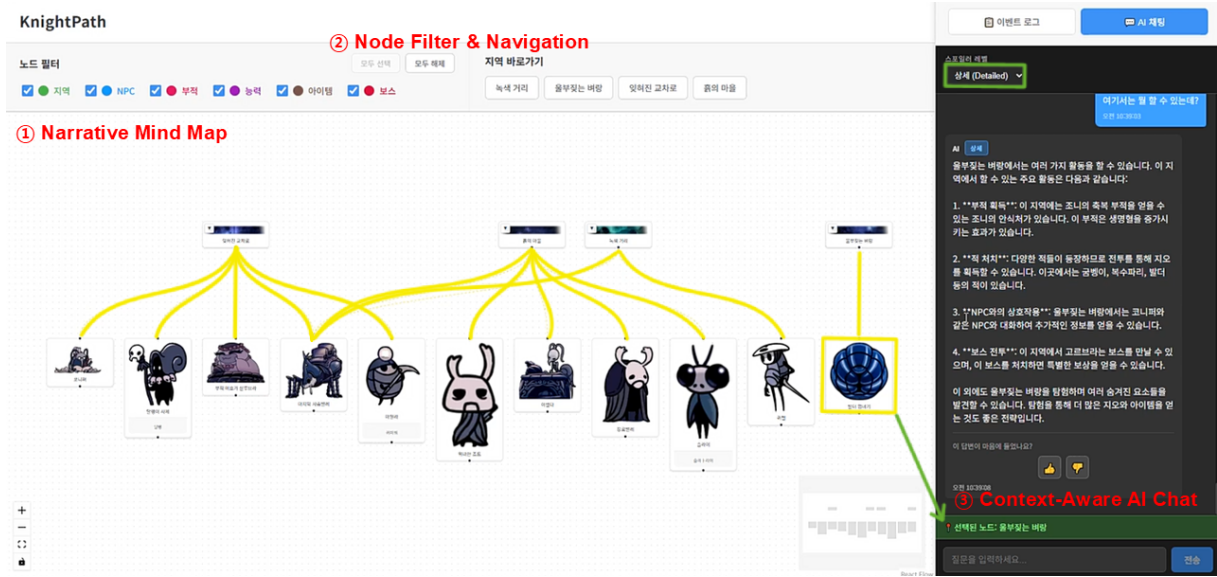


Figure 1: The Knight Path interface: (1) Narrative mind map organized by location clusters; (2) Node filtering and navigation controls; (3) Context-aware AI chat with adjustable spoiler levels.

perceived usability [1]. To understand players' **search goals**, we employed a triggered verbal protocol for each **Information-Seeking Episode (ISE)**, which represents the period from pausing gameplay and moving the cursor to the secondary monitor until resuming play. For each ISE, participants explained their immediate search goal at the onset of an ISE, and after each ISE, they were asked a brief post-ISE question regarding the perceived usefulness of the obtained information.

Behavioral Logs and Video Analysis. To capture player behavior, we synchronized digital traces—including in-game state, web history, and AI system logs—with screen recordings and verbal protocols. This enabled the tracking of raw interaction data, such as search and gameplay durations alongside specific search terms and queries. This dataset, verified and supplemented through video analysis, provided the ground truth for identifying Actions and reconstructing Inquiry Chains in the subsequent coding process.

Screen recordings and logs were analyzed with the **ISE** as the primary unit. Within each ISE, sub-activities were recorded as **Actions**, defined as fundamental *search moves* [17]. Coding criteria for actions were tailored to each group's environment: for the control group, an action was defined by a change in media (e.g., Google to YouTube); for the experimental group, actions were categorized by the tool utilized: (1) *Mind Map Interaction* (active interaction), (2) *Re-seeking* (history scrolling), and (3) *Querying Process* (from input box click to AI response evaluation). To capture the structure of player reasoning, actions were grouped into an **Inquiry Chain** based on the consistency of search objectives within each participant. This approach aligns with the concept of a *search tactic* [17] and the *sense-making process* [13] of synthesizing data into *insights* [6]. Rather than rigid thematic coding, we identified Inquiry Chains by triangulating search terms, queries, and verbal protocols. Each chain started with a new search goal and ended upon its resolution

or a topic shift, integrating all iterative follow-ups and keyword changes into a single continuous process. This allowed for comparing interaction patterns and the total actions required to resolve narrative issues across conditions.

4 Results and Discussion

Due to the small and non-normal sample, the Wilcoxon rank-sum test was used for group comparisons. Survey metrics were aggregated as **means**, while behavioral logs were converted into **medians** per participant to avoid bias from multiple episodes. Effect sizes were reported via Cliff's δ for robust estimates independent of sample size.

4.1 Information-Seeking Efficiency and Behavioral Patterns (RQ1)

The Wilcoxon rank-sum test showed no significant differences in **total search duration** (control: 4.9 vs. experimental: 7.9 min) and **frequency of ISE** (6.0 vs. 9.0). While the overall search volume remained relatively consistent, participants noted the conventional web-based searching method was cumbersome and inefficient for finding desired content due to information overload, citing it as a primary limitation. Despite these inefficiencies, control participants did not attempt to use generic AI tools during the study. Feedback indicated this was due to concerns regarding reliability, the burden of explaining game contexts, and the **risk of encountering unfiltered or excessive information**, including potential spoilers. This underscores the necessity of a context-aware filtering system.

Behavioral logs further supported these findings. The experimental group achieved significantly lower and more consistent **actions per ISE** ($Mdn : 1.0, SD : 0.38$ vs. $3.0, 1.19; p = .027, \delta = 0.653$) and **per inquiry chain** ($1.5, 0.41$ vs. $2.5, 1.62; p = .028, \delta = 0.71$). These results may be interpreted as reflecting the unstable nature

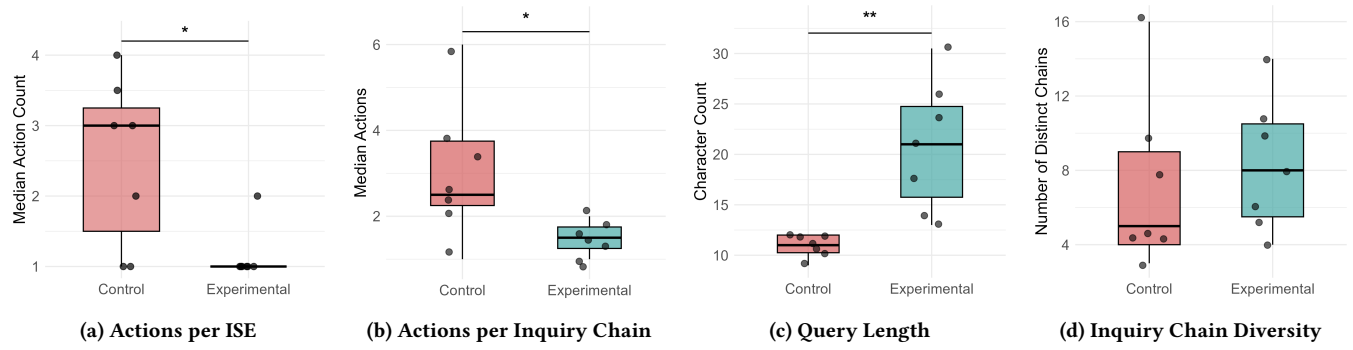


Figure 2: Behavioral metrics for information seeking efficiency and exploration quality: (a–b) frequency of actions per episode and inquiry chain, (c) query length, and (d) inquiry chain diversity.

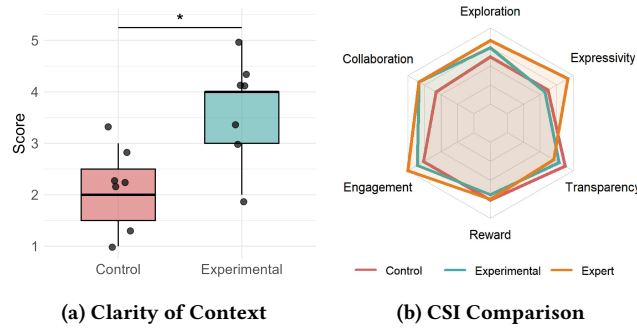


Figure 3: Subjective results: (a) significantly higher contextual clarity in the experimental group; (b) CSI comparison across all groups.

of conventional web-based searching, where costs fluctuate based on individual skill, whereas the experimental group demonstrates a more uniform and efficient pattern. The significantly longer **query length** in the experimental group ($Mdn : 11$ vs. 21 characters; $p = .002$) reflected a shift toward natural language interaction. Participants noted that this allowed them to address trivial or specific curiosities (P11, P18) as intuitively as *asking a peer* (P10, P16). Although the increase in **inquiry chain diversity** was not statistically significant (Mdn ; control: 5 vs. experimental: 8), these patterns suggest that natural language inputs may have broadened the accessibility of searching for a more diverse range of narrative issues by lowering the cognitive barrier to expressing complex queries.

4.2 Narrative Experience and System Evaluation (RQ2)

Subjective metrics confirmed that the system improved sense-making without compromising core gameplay; no significant differences were found in the mean scores of **post-interest change** (both $M = 4.14$) or **immersion** (control: 4.71 vs. experimental: 4.57). However, qualitative feedback revealed a distinct contrast in interest and enjoyment. While control group responses were diverse, all except for one participant in the experimental group reported a

positive influence. This was largely due to the mitigation of frustration, as P9 noted that receiving direction helped maintain interest in progression, and P17 and P18 mentioned that it converted the urge to quit into a sense of achievement. Additionally, P10 stated that spoiler-aware guidance was effective for sustaining curiosity.

Clarity of context was significantly higher in the experimental group than in the control ($M : 2.00$ vs. 3.57; $p = .012$, $\delta = 0.667$), effectively mitigating the ambiguity of conventional web-based searches. System usability was further validated by a mean SUS score of 80.36, reflecting high usability. Qualitative feedback highlighted that the narrative mind map aided memory organization via image-based nodes. Meanwhile, the AI assistant was valued for reducing search time and maintaining play flow. Although the AI was perceived as less definitive regarding spatial navigation (e.g., coordinates), its utility in facilitating next steps for game progression was highly regarded.

Regarding the **CSI**, while no significant differences were found in the mean scores between the control and experimental groups, specific subscales revealed meaningful trends. Both the experimental and expert groups reported higher **Collaboration** scores ($M = 6.00$, $SD : 2.37$) than the control group ($M = 4.43$, $SD : 0.82$). Qualitative feedback suggests participants viewed the system as a supportive partner. P18 (experimental) mentioned using the system as a *helper* for specific queries, while P14 (expert) described it as a *reliable assistant* standing by their side. Notably, the **expert** group exhibited significantly higher **Expressiveness** than the experimental group ($M = 6.50$ vs. 4.43; $p = .025$, $\delta = 0.857$). This aligns with qualitative feedback revealing a mastery-based shift in usage patterns. While novices sought aid for immediate progression, P13 noted experts leverage the system to *look into the story more*. This was corroborated by P14, who stated *asking about story* was the most helpful aspect of the system. These findings suggest that Knight Path’s expressive utility is maximized when users transition from functional progression to active narrative interpretation.

4.3 Spoiler Awareness

In this study, spoiler sensitivity was categorized based on pre-study survey responses, distinguishing between participants who are indifferent to spoilers and those who want to avoid them as much

as possible or strictly. Experimental group participants could utilize **spoiler level controls** to manage varying sensitivities. The spoiler-insensitive group tended to use *Detailed* or *High* levels for efficiency, with P8 stating they *wanted to know quickly* to move on while acknowledging the necessity of tiers for diverse preferences. Conversely, the spoiler-sensitive group engaged with a broader range of levels to manage narrative tension, most frequently selecting *Medium*. P10 and P16 mentioned that this tiered guidance allowed them to *maintain interest* by shielding discovery from premature disclosure. Qualitative interviews revealed that regardless of sensitivity, participants' **subjective evaluations** of information gaps varied significantly. These differing opinions stemmed from individual spoiler standards: P5, P12, and P13 mentioned primarily fearing narrative twists, while P6, P10, and P11 stated that mechanical solutions or losing the *fun of finding things* were equally disruptive. Such insights suggest that structural tiers require further personalization to balance unique spoiler criteria with the desired depth of assistance.

5 Conclusion

This research presented *Knight Path*, an adaptive information support system designed to bridge the gap between navigational aid and interpretive freedom in narrative-driven games. Our evaluation demonstrated that the system improves information-seeking efficiency and makes narrative context easier to grasp, all while maintaining immersion levels comparable to conventional searching. Notably, the system's utility shifts from a functional progression aid for novices to a tool for deep narrative interpretation as player mastery increases. This versatility, supported by a tiered spoiler control mechanism, allows players to manage narrative uncertainty without the risk of accidental exposure, effectively balancing essential guidance with the preservation of discovery.

Despite these contributions, the study is constrained by its small sample size ($N = 18$) and the highly subjective nature of spoiler standards. Qualitative interviews revealed that player sensitivity is deeply tied to specific types of disclosure, suggesting the potential for more refined approaches to personalized filtering and further validation in larger-scale studies. While evaluated in *Hollow Knight*'s fragmented narrative, the proposed approach warrants further investigation into its applicability to other narrative-driven games where players reconstruct story meaning from distributed elements. Additionally, future work could explore the development of algorithms capable of dynamically adjusting information granularity based on a player's unique narrative values and cognitive thresholds. Such a direction may further bridge the gap between support and exploration, potentially facilitating more expressive and engaging narrative experiences in increasingly complex game environments.

References

- [1] Aaron Bangor, Philip T. Kortum, and James T. Miller. 2008. An Empirical Evaluation of the System Usability Scale. *International Journal of Human-Computer Interaction* 24, 6 (2008), 574–594. doi:10.1080/10447310802205776
- [2] Erin A. Carroll, Celine Latulipe, Richard Au, and Elizabeth F. Churchill. 2009. The Creativity Support Index. In *CHI '09 Extended Abstracts on Human Factors in Computing Systems*. 4009–4014. doi:10.1145/1520340.1520609
- [3] Tobias Drey, Fabian Fischbach, Pascal Jansen, Julian Frommel, Michael Rietzler, and Enrico Rukzio. 2021. To Be or Not to Be Stuck, or Is It a Continuum? A Systematic Literature Review on the Concept of Being Stuck in Games. *Proceedings of the ACM on Human-Computer Interaction* 5, CHI PLAY, Article 229 (2021). doi:10.1145/3474656
- [4] Clara Fernández-Vara. 2011. Game Spaces Speak Volumes: Indexical Storytelling. In *Proceedings of the 2011 DiGRA International Conference: Think Design Play*. Digital Games Research Association (DiGRA).
- [5] Roberto Gallotta, Graham Todd, Marvin Zammit, Sam Earle, Antonios Liapis, Julian Togelius, and Georgios N. Yannakakis. 2024. Large Language Models and Games: A Survey and Roadmap. *IEEE Transactions on Games* 16, 4 (2024), 394–411. doi:10.1109/TG.2024.3461510
- [6] Oliver Gstöttenbauer, Claire Dormann, and Günter Wallner. 2025. Player-Centric Data Visualization for *Metrodvania* Games: Insights and Use Cases: The Case of *Hollow Knight*. *Proceedings of the ACM on Human-Computer Interaction* 9, 6, Article GAMES011 (2025). doi:10.1145/3748606
- [7] Sabina Gumulak and Sheila Webber. 2011. Playing video games: learning and information literacy. *Aslib Proceedings* 63, 2/3 (2011), 241–255. doi:10.1108/00012531111135682
- [8] J. Tuomas Harviainen and Juho Hamari. 2015. Seek, share, or withhold: information trading in MMORPGs. *Journal of Documentation* 71, 6 (2015), 1119–1134. doi:10.1108/JD-09-2014-0135
- [9] Juho Kahila, Teemu Valtanen, Sonsoles López-Pernas, Mohammed Saqr, Henriikka Vartiainen, Sini Kahila, and Matti Tedre. 2025. A Typology of Metagamers: Identifying Player Types Based on Beyond the Game Activities. *Games and Culture* 20, 1 (2025), 38–58. doi:10.1177/15554120231187758
- [10] Erica Kleinman, Nikitha Preetham, Zhaoqing Teng, Andy Bryant, and Magy Seif El-Nasr. 2021. "What Happened Here?" A Taxonomy for User Interaction with Spatio-Temporal Game Data Visualization. *Proc. ACM Hum.-Comput. Interact.* 5, CHI PLAY, Article 260 (2021). doi:10.1145/3474687
- [11] Jungmin Lee, Seoyoung Yoon, Hwajin Shim, and Youngjae Yoo. 2025. Development of an LLM-Based Chatbot to Support Learnability in *Stardew Valley*: A Diary Study Approach. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Article 1070. doi:10.1145/3706598.3713310
- [12] NamuWiki contributors. [n. d.]. *Hollow Knight*. <https://namu.wiki/w/Hollow%20Knight>.
- [13] Peter Pirolli and Stuart Card. 2005. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. In *Proceedings of International Conference on Intelligence Analysis*, Vol. 5. 2–4.
- [14] Zeyu Tang and Ben Kirman. 2025. Exploring Curiosity in Games: A Framework and Questionnaire Study of Player Perspectives. *International Journal of Human-Computer Interaction* 41, 4 (2025), 2475–2490. doi:10.1080/10447318.2024.2325171
- [15] Günter Wallner, Marnix van Wijland, Regina Bernhaupt, and Simone Kriglstein. 2021. What Players Want: Information Needs of Players on Post-Game Visualizations. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Article 330. doi:10.1145/3411764.3445174
- [16] Xiyuan Wang, Yi-Fan Cao, Junjie Xiong, Sizhe Chen, Wenxuan Li, Junjie Zhang, and Quan Li. 2025. ClueCart: Supporting Game Story Interpretation and Narrative Inference from Fragmented Clues. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Article 410, 26 pages. doi:10.1145/3706598.3713381
- [17] Barbara M. Wildemuth. 2004. The effects of domain knowledge on search tactic formulation. *Journal of the American Society for Information Science and Technology* 55, 3 (2004), 246–258.